



PES

UNIVERSITY

CELEBRATING 50 YEARS

ALGORITHMS ANALYSIS AND DESIGN

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Computer Applications

ALGORITHMS ANALYSIS AND DESIGN

Introduction, Analysis Framework and Sorting Techniques

Sorting Algorithms – Decrease and Conquer

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ALGORITHMS ANALYSIS AND DESIGN

Insertion Sort

- Sort an array $A[0..n-1]$ using the decrease-by-one technique.
- Three alternatives
 1. Scan the sorted sub array from left to right until the first element greater than or equal to $A[n-1]$ is encountered.
 - Then insert $A[n-1]$ right before the element.



ALGORITHMS ANALYSIS AND DESIGN

Insertion Sort

2. Scan the sorted sub array from right to left until the first element smaller than or equal to $A[n-1]$ is encountered and then insert $A[n-1]$ right after that element.
- This method is well suited for sorted arrays and almost sorted arrays.
 - This resulting algorithm is called straight selection sort.



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Insertion Sort

3. The third alternative is to use binary search to find an appropriate position for $A[n-1]$ in the sorted portion of the array.
- The resulting algorithm is called binary insertion sort.



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Algorithm – InsertionSort($A[0..n-1]$)

- //Sorts a given array by insertion sort
- //Input: An array $A[0..n-1]$ of n orderable elements
- //output: Array $A[0..n-1]$ sorted in non-decreasing order



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Algorithm – InsertionSort($A[0..n-1]$)

- for $i \leftarrow 1$ to $n-1$ do
- $v \leftarrow A[i]$
- $j \leftarrow i-1$
- while $j \geq 0$ and $A[j] > v$ do
- $A[j+1] \leftarrow A[j]$
- $j \leftarrow j - 1$
- $A[j+1] \leftarrow v$



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Algorithm – InsertionSort($A[0..n-1]$) - Tracing

A=	4	1	5	2	8			n	5			
	0	1	2	3	4			n-1	4			
Pass No.	i	$i \leq n-1$	v	j	Col 1	Col 2	Col1 and Col 2	A				
					$j \geq 0$	$A[j] > v$						
								0	1	2	3	4
								4	1	5	2	8
1	1	$1 \leq 4$ (T)	1	0	$0 \geq 0$ (T)	$4 > 1$ (T)	T	4	4	5	2	8
				-1	$(-1) \geq 0$ (F)			1	4	5	2	8
2	2	$2 \leq 4$ (T)	5	1	$1 \geq 0$ (T)	$4 > 5$ (F)	F	1	4	5	2	8
3	3	$3 \leq 4$ (T)	2	2	$2 \geq 0$ (T)	$5 > 2$ (T)	T	1	4	5	5	8
				1	$1 \geq 0$ (T)	$4 > 2$ (T)	T	1	4	4	5	8
				0	$0 \geq 0$ (T)	$1 > 2$ (F)	F	1	2	4	5	8
4	4	$4 \leq 4$ (T)	8	3	$3 \geq 0$ (T)	$5 > 8$ (F)	F	1	2	4	5	8



ALGORITHMS ANALYSIS AND DESIGN

Algorithm – InsertionSort($A[0..n-1]$) - Tracing

Pass Number	Number of Comparisons
1	1
2	1
3	3
4	1



ALGORITHMS ANALYSIS AND DESIGN

Recap

- Insertion Sort
 - Algorithm
 - Tracing



THANK YOU

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